

Course

- Name: DIGITAL SIGNAL PROCESSING
- Code: FME3206/MLE5178
- Credits: 5
- Schedule:
 - 2 hrs lecture, Thursday. 4 pm, AE
 - 2 hrs lab, Mon. 2 pm (FM+FC + FCS), Thursday
 - 2 pm (CS1), 6 pm (CS2), room 215
- Material: available via [Ms Teams](#)



Tutor

- Name: Lázár, Zsolt I.
- Email: zsolt.lazar@ubbcluj.ro
- Location: main building, room 203.

Final assessment

- Exam (40%)
- Homeworks (25%) (TBD)
- Project (20%) (TBD)
- Presentation (15%)(TBD)

What is a signal?

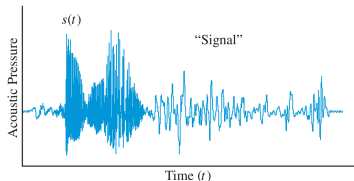
- flow of information
- any (physical) quantity that varies with time (or position). Mathematically, we describe a signal as a function of one or more independent variables;
- electrical signal received from a transducer: microphone, thermometer, accelerometer, antenna, etc.
- electrical signal that controls a process

- **Mathematical:** see also the figures on the next slide
 - Function $s(t)$ or $s(x)$
 - Sequence $s[n]$
- **Physical:** the storage, transmission, and processing of signals require their representation using physical media, e.g., voltage levels, magnetic domains on tape, optical domains on a CD

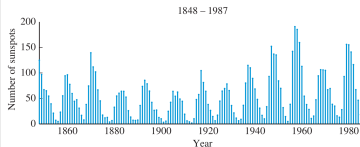
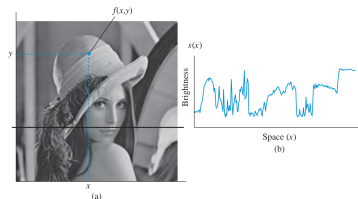
Types of signals

- **Temporal:** voltage, blood pressure, etc.
- **Spatial:** images, lunar landscape, etc.
- **Continuous-time:** voltage, current, temperature, speed, etc.
- **Discrete-time:** daily minimum/maximum temperature, lap intervals in races, sampled continuous signals, etc.
- **Analog:** amplitude values from a continuous range
- **Digital:** amplitude values from a discrete set. Converted from analog signal by quantization (to a discrete set of possible values).

Electronics (unlike optics) can only deal easily with time-dependent signals, therefore spatial signals, such as images, are typically first converted into a time signal with a scanning process (TV, fax, etc.).



Example of a recording of speech. The time waveform shows the variation of acoustic pressure as a function $s(t)$ of time for the word "signal."



- continuous-time signal:

$s(t) : \mathbb{R} \rightarrow \mathbb{R} \in C(\mathbb{R})$ continuous function

t or $x \in \mathbb{R}$ - *independent variable* $\rightarrow$ time

$s \in \mathbb{R}$ - *dependent variable* $\rightarrow$ amplitude

Also referred to as analog signal (sloppy!).

- discrete-time signal (sequence):

$s[n] : \mathbb{Z} \rightarrow \mathbb{R} \equiv s(t)|_{t=nT} = s(nT)$ - uniform *sampling*

T *sampling period*, $F_s \equiv 1/T$ *sampling frequency* or *sampling rate*

- digital signal:

$s_d[n] : \mathbb{Z} \rightarrow \mathbb{X}$, $\mathbb{X}$ - discrete set, e.g., $\mathbb{Z}$

Obtainable by *quantizing* the set of amplitude values. Also sloppy!

All combinations of (continuous-time, discrete-time) $\times$ (analog, digital) are possible.

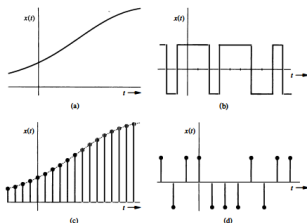
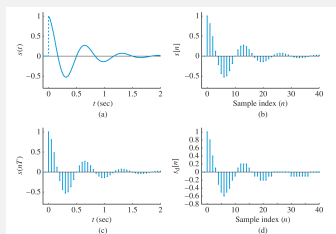


Figure 1.11 Examples of signals: (a) analog, continuous time, (b) digital, continuous time, (c) analog, discrete time, and (d) digital, discrete time.

Example



continuous-time

$$s(t) = \begin{cases} e^{-2t} \cos(3\pi t), & t \geq 0 \\ 0, & t < 0 \end{cases}$$

discrete-time $T = 0.1$ s

$$s[n] = s(nT) = \begin{cases} e^{-0.2n} \cos(0.3\pi n), & n \geq 0 \\ 0, & n < 0 \end{cases}$$

digital $\Delta = 0.1$

$$s_d = \left\lceil \frac{s}{\Delta} \right\rceil \Delta \in \{-0.6, -0.5, \dots, 0.9, 1\}$$

17 possible values of s_d

Physical representation

- **analog:** voltage or current that is proportional to the value of the quantity. Requires using the physical characteristics of the storage medium to create “continuous analogies” for time and amplitude.

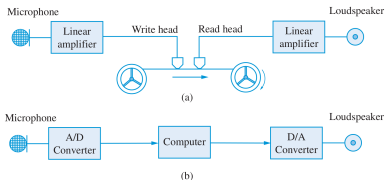
Example: analog tape recording

time $\leftarrow$ increasing linear distance along the tape;

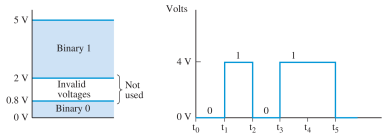
amplitude $\leftarrow$ magnetic field of the tape.

Advantage: theoretically, infinite resolution, i.e., high fidelity.

Drawbacks: distortion of time (speed of tape) and of amplitude (change in the state/reading of the magnetic fields)



- **digital:** combination of ON/OFF pulses corresponding to the digits of a binary number



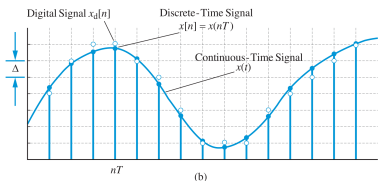
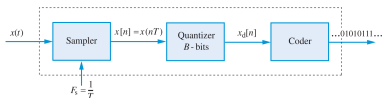
- **System** is a process where a signal called *input* is transformed into another signal called *output*.
- **Continuous-time system** is a system which transforms a continuous-time input signal $x(t)$ into a continuous-time output signal $y(t)$

$$x(t) \xrightarrow{\mathcal{H}} y(t), \quad \text{e.g.} \quad y(t) = \int_{-\infty}^t x(\tau) d\tau.$$

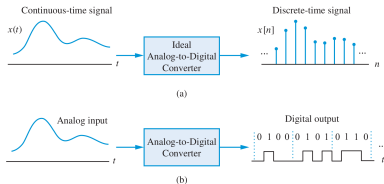
- **Discrete-time system** is a system which transforms a discrete-time input signal $x[n]$ into a discrete-time output signal $y[n]$

$$x[n] \xrightarrow{\mathcal{H}} y[n] \quad \text{or} \quad y[n] = \mathcal{H}\{x[n]\}, \quad \text{e.g.} \quad y[n] = \sum_{k=-\infty}^n x[k].$$

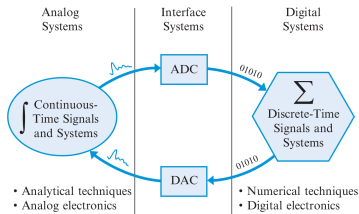
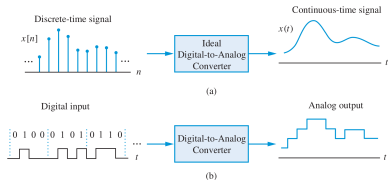
- **Interface system** is a system that provides the interface between analog and digital signals.



• Analog-to-digital conversion



• Digital-to-analog conversion



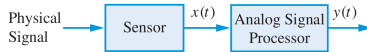
- **Definition:** Concerned with the acquisition, representation, manipulation, and transformation of signals.
- **Objectives:**
 - improve the quality of a signal
 - undo distortions contributed by a transmission channel
 - compensate for sensor deficiencies
 - amplify or filter out embedded information
 - extract useful information from a signal
 - detect patterns
 - separate previously combined signals
 - prepare signals for storage and transmission.
 - prevent interference with other signals sharing a medium

To do so, we also need

- methods to measure, characterise, model and simulate transmission channels
- mathematical tools that split common channels and transformations into easily manipulated building blocks

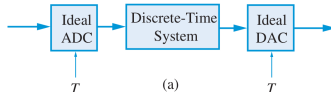
- **Analog signal processing**

- conversion of analog signals into electrical signals by special transducers or sensors
- conditioning of sensor output, e.g., amplification
- processing by analog electrical and electronic circuits.

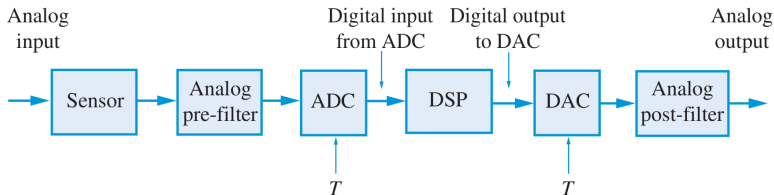


- **Digital signal processing**

- started in the 1960's
- representation of analog signals by sequences of numbers,
- processing of these sequences by numerical computation techniques
- conversion into analog signals.



Ideal (infinite resolution) signal processing system



Also referred to as *mixed signal processing* ← both analog and digital components are implemented on the same integrated circuit.

- **communication systems** modulation/demodulation, channel equalization, echo cancellation
- **astronomy** VLBI, speckle interferometry
- **consumer electronics** perceptual coding of audio and video on DVDs, speech synthesis, speech recognition
- **experimental physics** sensor-data evaluation
- **aviation** radar, radio navigation
- **music** synthetic instruments, audio effects, noise reduction
- **security** steganography, digital watermarking, biometric identification, surveillance systems, signals intelligence, electronic warfare
- **medical diagnostics** magnetic-resonance and ultrasonic imaging, computer tomography, ECG, EEG, MEG, AED, audiology
- **geophysics** seismology, oil exploration
- **engineering** control systems, feature extraction for pattern recognition

- **absolutely summable** $\Leftrightarrow \sum_{n=-\infty}^{\infty} |x[n]| < \infty$
- **square summable** $\Leftrightarrow \sum_{n=-\infty}^{\infty} |x[n]|^2 < \infty$
- **periodic** $\Leftrightarrow \exists k > 0 : \forall n \in \mathbb{Z} : x[n] = x[n + k]$

A square-summable sequence is also called an **energy signal**:

$$|x[n]|^2 \quad - \quad \text{power} \qquad \sum_{n=-\infty}^{\infty} |x[n]|^2 \quad - \quad \text{energy}$$

A non-square-summable sequence is a **power signal** if its average power exists:

$$\lim_{k \rightarrow \infty} \frac{1}{1 + 2k} \sum_{n=-k}^k |x[n]|^2 < \infty .$$

Examples

Absolutely summable

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty ? \quad \text{Example: } x[n] = \left(\frac{1}{3}\right)^{|n|} \quad \text{Counterexample: } x[n] = \frac{1}{|n| + 1}$$

Square summable

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 < \infty ? \quad \text{Example: } x[n] = \frac{1}{|n| + 1} \quad \text{Counterexample: } x[n] = \frac{1}{\sqrt{|n| + 1}}$$

Periodic

$$\exists k > 0 : \forall n \in \mathbb{Z} : x[n] = x[n + k] ? \quad \text{Example: } x[n] = (-1)^n \quad \text{Counterexample: } x[n] = n$$

Power signal

$$\lim_{k \rightarrow \infty} \frac{1}{2k + 1} \sum_{n=-k}^k |x[n]|^2 < \infty ? \quad \text{Example: } x[n] = \sin\left(\frac{\pi}{4} n\right) \quad \text{Counterexample: } x[n] = n^2$$

Types of discrete systems

- **causal** - only depends on previous input values:

$$y[n] = f(x[n], x[n-1], x[n-2], \dots) .$$

- **memory-less** - only depends on the current input value:

$$y[n] = f(x[n]) .$$

- **delay** - shifts a sequence in time:

$$y[n] = x[n-d] .$$

- **time-invariant** $\mathcal{H}$ for any d if

$$y[n] = \mathcal{H}(x[n]) \quad \Leftrightarrow \quad y[n-d] = \mathcal{H}(x[n-d]) .$$

- **linear** $\mathcal{H}$ if

$$\mathcal{H}(a_1 x_1[n] + a_2 x_2[n]) = a_1 \mathcal{H}(x_1[n]) + a_2 \mathcal{H}(x_2[n]) .$$

- **stable** if

$$|x[n]| < M_x < \infty \quad \Rightarrow \quad |y[n]| < M_y < \infty$$

- **...linearity** (principle of superposition):
 1. *homogeneity* or *scaling*: if $x \rightarrow x' \equiv \alpha x$, check that $y = \mathcal{H}(x) \rightarrow y' = \mathcal{H}(x') = \alpha y$
 2. *additivity*: if $y_1 = \mathcal{H}(x_1)$ and $y_2 = \mathcal{H}(x_2)$, check that $y_1 + y_2 = \mathcal{H}(x_1) + \mathcal{H}(x_2)$

Note that if $\mathcal{H}(x)$ is a linear combination of x , e.g., $2x[n-1] - \frac{1}{2}x[n] + x[n+1]$ then linearity is satisfied.
- **...causality**: if the right hand side of $y[n] = \mathcal{H}(x[n])$ does not involve $x[k]$ elements with $k > n$, e.g., $y[n] = x[n] - x[n-1]$ is causal while $y[n] = (x[n] + x[n+1])/2$ is not.
- **...memory-less**: if the rhs of $y[n] = \mathcal{H}(x[n])$ does not involve $x[k]$ elements with $k \neq n$, e.g., $y[n] = x[n]^2$ is memory-less while $y[n] = x[n] - x[n-1]$ or $y[n] = (x[n] + x[n+1])/2$ are not.
- **...time-invariant**: if on the rhs of $y[n] = \mathcal{H}(x)$, $x[\text{some index}]$ is replaced by $x[\text{some index} + d]$, check that $\mathcal{H}(x[\text{some index} + d]) = y[n + d]$. E.g. $y[n] + x[n]u[n] \rightarrow x[n+1]u[n] \neq y[n+1] = x[n+1]u[n+1]$.
Systems of type $x[n]f(n)$ are not time-invariant unless $f = \text{const.}$

Examples

Causal, linear, time-invariant systems with memory

- accumulator system:

$$y[n] = \sum_{k=-\infty}^n x[k] .$$

- backward difference system:

$$y[n] = x[n] - x[n-1] .$$

- M point moving average system:

$$\begin{aligned} y[n] &= \frac{1}{M} \sum_{k=0}^{M-1} x[n-k] = \\ &= \frac{x[n] + x[n-1] + \dots + x[n-M+1]}{M} . \end{aligned}$$

- exponential averaging system:

$$y[n] = \alpha x[n] + (1-\alpha)y[n-1] = \alpha \sum_{k=0}^{\infty} (1-\alpha)^k x[n-k] .$$

Nonlinear, time-invariant, memory-less systems

-

$$y[n] = x[n]^2 ,$$

-

$$y[n] = \log_2 x[n] ,$$

-

$$y[n] = \max \{ \min \{ \lfloor 256x[n] \rfloor , 255 \} , 0 \}$$

Linear, not time-invariant systems

-

$$y[n] = \begin{cases} x[n] , & n \geq 0 \\ 0 , & n < 0 \end{cases} = x[n]u[n] .$$

-

$$y[n] = x[\lfloor n/4 \rfloor] .$$

-

$$y[n] = x_n \operatorname{Re} \left(e^{i\omega n} \right)$$

Linear, time-invariant, non-causal systems

- two point moving average system:

$$y[n] = \frac{1}{2} (x[n-1] + x[n+1]) .$$

-

$$y[n] = \sum_{k=-9}^9 x[n+k] \frac{\sin(\pi k \omega)}{2\pi k \omega} \left[1 + \cos \left(\frac{\pi k}{10} \right) \right] .$$

Exercises

What type of discrete system ((non-)linear, (non-)time-invariant, (non-)causal, memory-less, etc.) is

① $y[n] = |x[n]|$

② $y[n] = -x[n-1] + 2x[n] - x[n+1]$

③ $y[n] = \prod_{k=0}^8 x[n-k]$

④ $y[n] = \frac{1}{2} (x[2n] + x[2n+1])$

⑤ $y[n] = \frac{3x[n-1] + x[n-2]}{x[n-3]}$

⑥ $y[n] = x[n]e^{n/14}$

⑦ $y[n] = x[n]u[n]$

⑧ $y[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[k-n+2]$